A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

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ABSTRACT. This paper proposes a minimal core of necessary conditions for registration, the basic form of any epistemic relation: conditions without which nothing could be perceived, or registered in any way, whatever the domain in which the relation is realized. The core rests on two notions that it does not analyse, a field's bearing a contrast and one contrast's holding partly in virtue of another, and it admits a single commitment: receptive dependence, by which a contrast borne by a field holds at least partly in virtue of a contrast borne by the field under consideration. Two further conditions follow and are not admitted separately: that the field under consideration bears the contrast registered, and that the contrast which depends is a second contrast, whose bearer is a field in the same sense. Contrasts are throughout between actual terms, never between alternatives, so that the core requires no repertoire of possibilities on either side. On this basis, registration is defined in a strictly structural sense that presupposes neither understanding, nor representation, nor order. The conditions are formal: they presuppose no spatial, temporal, or physical structure. They are claimed to be necessary, not sufficient, and the necessity is conditional: whatever is registered satisfies them. The core is a schema: it states what any dependence must satisfy to count as registration, and leaves to each domain what the dependence consists in. The paper further shows that target and receiver are positions in the relation of dependence rather than natures of what occupies them, states as a condition of method that specifying an articulation selects it without creating it, and shows that acquisition and retention require an ordering which the core does not contain and whose kind it leaves open.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper addresses a prior question: what must be admitted for any epistemic relation to be possible at all—for anything to be perceived, or registered in any way. Its answer is a set of necessary conditions, not a sufficient account. It

identifies what any theory of perception, belief, or knowledge presupposes, and leaves to such theories what makes a relation epistemic in the full sense. The assessment of judgement, in particular, lies outside its scope.

The guiding premise is that, in its basic form, prior to concepts and descriptions, an epistemic relation is always a relation to something in particular. What is perceived, or registered in any way, is a difference: X as it differs, in some respect, from an actual other term. And the registering must depend on that difference; otherwise what is perceived or registered would not be about it. The premise concerns this basic form only. A claim may be about something by description, about a future event, or about a mathematical object, and the premise does not cover these cases; whether they are built from the basic form is left open (Section 8).

The necessity claimed is accordingly conditional. Necessarily, whatever is registered, of whatever kind, satisfies the conditions stated below, because they follow from the guiding premise and from the definitions. It is not claimed that registration occurs, nor that the conditions hold of anything apart from registration.

The conditions are formal. They are stated without reference to space, time, matter, or causation, and they are meant to hold whatever the domain in which registration is realized, whether perceptual, physical, or linguistic. Examples drawn from particular domains show how the conditions are met there; they do not fix what the conditions are. For the same reason, necessity is claimed only where it can be argued for, and features of the presentation that are conventions rather than commitments are marked as such.

The reconstruction proceeds by minimal commitment. Each element is introduced only at the point where the account cannot proceed without it, and each is accompanied by a statement of what it does not establish. One methodological caution applies throughout. The exposition necessarily employs ordinary logical resources—identity, distinction, relation, and inference—but using them to state the commitments does not attribute them to what the commitments describe, and in particular not to a knowing subject. A distinction drawn in the exposition is not thereby a distinction drawn by its object.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither and concerns only the dependence of one contrast on another.

describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from each in three respects. It separates the contrast that is registered, the contrast that registers it, and the dependence that connects them, and it admits only the last as a commitment. It states that dependence between contrasts of actual terms, without probabilities, causes, or alternatives on either side, so that it applies beyond the physical domain and requires no repertoire of possibilities. And it distinguishes the direction of dependence from any order of occurrences, leaving open what kind of order further epistemic operations require.

Section 2 introduces the notion of a field's bearing a contrast and the condition of method under which the reconstruction proceeds. Section 3 defines articulations and their realization, and states two consequences. Section 4 introduces receptive dependence, the single commitment, and the roles it defines; Section 5 draws from it the second field. Section 6 defines registration on that basis and states what it does not include, and Section 7 shows that the resulting structure contains no order. Section 8 summarizes the elements of the account, what they exclude, their limitations, and the questions they leave open.

2. Fields and Contrasts

The account uses one notion without analysing it: that something *bears a contrast* in some respect. Whatever bears a contrast is, insofar as it does, a *field*. To be a field is thus an aspect, not a kind of thing: the term implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Let F designate the field under consideration. The designation fixes the scope of the inquiry. It does not establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

That F bears a contrast is not a modal claim. It does not say that F can be differentiated, or that it admits differentiations which it does not have; it says that a difference is there. Nothing is thereby said about which contrasts F bears or of what form, about whether some differentiation of F is uniquely privileged, or about whether what differs would be objects, qualities, regions, values, or something else. Nor is anything implied about change or about any order. Possibilities and capacities of a field are not thereby denied. They are not part of the core, and if a later stage of the theory needs them, they must enter as commitments of their own, stated as such.

The reconstruction is subject to one condition. It is a condition of method,

not a claim of the core.

SCOPE OF THE RECONSTRUCTION

The reconstruction articulates only fields taken to bear the contrasts it specifies.

The reason is that the reconstruction concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description. This is a necessary condition for the route of explanation pursued here, not a result reached by it. The account does not argue against the view that all differentiation is supplied by descriptions. It states what that view would cost: if it held, nothing would be registered in the sense of the guiding premise. No knowing subject is involved at this point either: that F bears a contrast does not imply that anything differentiates it.

3. Articulation

A particular differentiation of F will be called an *articulation* and written

$$\langle a, \delta, b \rangle_F,$$

where a and b are its *terms* and δ is its *respect*. Its *contrast* is the difference of a and b in that respect; two articulations may share a respect and have different contrasts. Two levels must be kept apart. A *position* is a place in a contrast that a field bears. A *specified term* is what a description picks out, and it may pick out nothing. An articulation is *specified* when its terms and respect are fixed, and it is the account, not F, that specifies it. It is *realized* when its specified terms pick out positions of a contrast that F bears in the respect δ. Without the two levels, every articulation would be realized by definition.

The terms of a realized articulation are both actual. They are not alternatives of which one is there rather than the other, and the contrast is not a choice between them; it is the difference between two terms that are both there. The schema has two terms for simplicity; the account requires only that an articulation have at least two. Nor does the schema presuppose that a and b exist prior to the contrast as independent objects; it marks positions within it.[2]

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ expresses the differentiation itself. Second, b

2. The refusal to ground a difference in terms identified before it has a precedent in Deleuze's critique of the subordination of difference to identity (Deleuze 1994, ch. 1). Deleuze also rejects the possible as a mere image of the real; the present account shares the refusal of a stock of possibilities.

need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other. Figure 1 sets the two constructions side by side.

Specification chooses the respect and selects a contrast borne in it. That it does not create the contrast is a stipulation of the route stated in Section 2, not something the account proves. Whether F realizes the specified articulation, and whether anything depends on it, are not settled by the specification, which can therefore fail. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

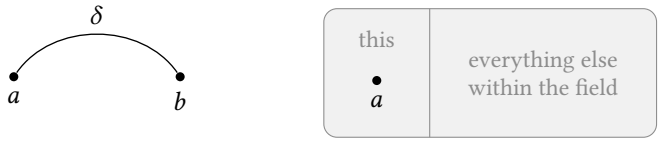
The condition of Section 2 accordingly leaves out of the reconstruction any respect whose holding depends, even partly, on the specification or its notation: on the order, the labels, or the grouping of the terms. Suppose that F bears a contrast in the respect δ, and let δ^* be the respect: δ, and a is written first. F does not bear a contrast in δ^*, because part of what makes δ^* hold is not in F. A differentiation of this mixed kind is not excluded wholesale. The contrast that F bears, in δ, is within the scope of the reconstruction; the remainder is not.

Two consequences follow at once. The first, recorded as (D), is that every field in which a registered articulation is realized bears that articulation's contrast. This is not a commitment. It follows from the definition of a realized articulation, and indeed from that of a field, and it holds of any realized articulation, registered or not. It establishes neither which contrasts a field bears, nor that any field bears a contrast apart from registration, nor any modal claim about F. The second concerns a lone term. A contrast has at least two positions, both actual. What differed from no actual other term, in any respect, would be a position in no contrast, and nothing in it could be registered.

These qualifications mark the first limit of the reconstruction. That a field bears contrasts does not fix which articulation an account will specify. An account of any particular case must therefore specify the articulation it concerns, and it may fail to pick out anything. Moreover, a realized articulation remains an articulation of F, and nothing established so far implies that it is registered anywhere.

4. Receptive Dependence

That F bears a contrast, and that some other contrast is borne somewhere, leaves the two unconnected. The guiding premise requires more: the registering of a difference must depend on that difference. A relational commitment is therefore required, and it is the only commitment of the core. Let C designate *receptive*



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark positions, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

dependence.

(C) RECEPTIVE DEPENDENCE

A contrast K', borne by a field, holds at least partly in virtue of a contrast K borne by F.

(C) relates two particular contrasts. An account states an instance of it by specifying an articulation whose contrast is K; the specification belongs to the account, and the dependence does not wait on it.

The phrase "in virtue of" is the second notion that the account does not analyse. K' holds because of K, and not conversely or merely alongside it. The phrase is not meant to identify C with metaphysical ground; it borrows only two features that the literature on ground has made familiar, namely that dependence of this kind is asymmetric and that it relates what actually holds, without appeal to what would hold otherwise (Fine 2012). (C) is accordingly not a counterfactual claim. It does not say what K' would be if F were otherwise; it relates two contrasts that both hold.[3]

Dependence must be distinguished from two relations that are symmetric. Two contrasts may resemble one another without either registering the other,[4] and the terms of one may be paired with the terms of the other, a with r and b with r', without either contrast holding because of the other (Figure 2). Such a pairing can be read in either direction. (C) requires no pairing of terms. It relates one contrast to another, and which position of K' answers to which position of

3. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997). Neither difficulty affects (C).

4. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.

K is not part of it. Where a pairing is needed, as it may be for registering which way F is (Section 6), it must enter as a further commitment, stated as such.

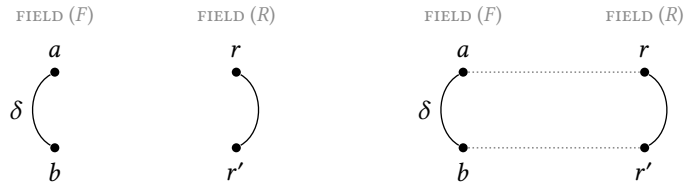
Within the reconstruction the dependence is asymmetric in a strict sense: if K' holds at least partly in virtue of K, then K does not hold, even partly, in virtue of K'. The ground is the condition of Section 2. The reconstruction concerns a differentiation attributable to its target. A contrast made in part by a description lies outside it, and so, for the same reason, does a contrast made in part by its own registration. The strict asymmetry is therefore a matter of scope, like the exclusion of notational respects, and not a result: the guiding premise says that the registering must depend on the difference, not that the difference must be independent of the registering. Within that scope, no contrast holds in virtue of itself. The asymmetry holds between contrasts, not between fields: two fields may each depend on the other through different contrasts. In particular, K' may hold partly in virtue of K while a further contrast K'' of F holds partly in virtue of K', and an interaction that produces differences in F can be described in this way. Whether the dependence is transitive is left open. If it is not, loops through a third contrast are not excluded; if it is, the dependence is a strict partial order among contrasts.

The dependence may be partial. K' holds at least partly in virtue of K, and whatever else it holds in virtue of may lie in the field that bears K'.[5] The point is captured in Bateson's definition of information as "a difference which makes a difference" (Bateson 1972), anticipated in MacKay's "a distinction which makes a difference" (MacKay 1969), which compresses into one phrase what the present account keeps apart: the first difference is K, the second is K', and the making is the dependence C. For the same reason, the content of (C) should not be concealed behind appeals to "systematic" or "relevant" coupling, which import what the argument has not supplied, namely a standard of relevance or a regularity across cases.

Whether (C) holds in an instance is fixed neither by descriptions of the two contrasts nor by what would hold otherwise. Two instances alike in their contrasts, and in any correspondence between their terms, may therefore differ in whether (C) holds: one is registration, the other mere correspondence. Whether (C) in an instance can depend on other actual instances, as in a regularity across cases, is left open.

(C) does not single out one registered contrast. If K' holds partly in virtue of K and partly in virtue of another contrast K^*, it stands in C to both; and if the dependence is transitive, it stands in C to whatever K in turn depends on. At this level a contrast registers every contrast it holds at least partly in virtue of. To narrow this to one is not a condition of registration but a further achievement, for which a later stage of the theory must supply the means.

5. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).



(a) resemblance (b) correspondence

FIGURE 2. Two relations that fall short of receptive dependence. (a) Resemblance: two contrasts share a form, but neither registers the other. (b) Correspondence: a is paired with r, and b with r'. The pairing is drawn without direction, since it can be read either way. Neither panel establishes (C), and (C) requires neither.

The asymmetry of C defines two roles. In an instance of C, the contrast depended upon is the *registered* contrast and the contrast that depends is the *registering* contrast. The field that bears the first is the *target* in that instance, and the field that bears the second is the *receiver*. Roles belong to contrasts first, and to fields through them: one field may bear both contrasts, and is then target and receiver in the same instance. Roles are thus positions in a relation, not natures of what occupies them. The statement of (C) does not use the notion of a receiver: C is introduced as an asymmetric relation between contrasts, and the roles are read off it.

Two limitations of (C) should be noted. First, the dependence is admitted, not derived from the existence of contrasts; a relation cannot be obtained merely by describing what it relates. C is introduced because the guiding premise requires a connection that no description of the two contrasts supplies. Second, the dependence is primitive in the core. It is characterized as relating what actually holds and, within the scope of the reconstruction, as asymmetric, and it is not analysed further. What it consists in within a particular domain is a question for that domain: in physical domains it may be causal, in others it may be logical or constitutive. The core is in this sense a schema. It states what any dependence must satisfy to count as registration, and each domain supplies the dependence; it requires that there be such a dependence, not that it be of any one kind.

5. The Receiving Field

(C) relates two contrasts, and they are two: since no contrast holds in virtue of itself, K' is distinct from K. The second consequence follows.

(R) SECOND FIELD

In an instance of C, the contrast that depends is distinct from the contrast depended upon. What bears it is, insofar as it does, a field; R designates

it.

(R) is not a further commitment. It follows from (C), from its asymmetry, and from what a field is.[6] R is a field in exactly the sense in which F is one: whatever bears a contrast, insofar as it does. Bearing a contrast is not a property peculiar to receivers. What holds of the target therefore holds of R: the contrast concerned is attributable to the field that bears it, and it is the account that picks it out. R itself distinguishes nothing. In particular, it does not distinguish registration from its absence: that K' holds partly in virtue of K is a fact about the two contrasts, not something that R notes.

R is not a person, a conscious subject, or an independently bounded system. Nothing follows about whether R and F are distinct fields, parts of a larger one, or a single field bearing two contrasts, since a distinction of contrasts does not entail a distinction of substances. A field may register one of its own contrasts, but only through a second contrast.

The positions of K' will be called *conditions* of R and written $r, r', \ldots$. As with the terms of a realized articulation, they are both actual: r and r' are not alternatives between which R is poised, but distinct conditions that are both there. Nothing requires that K' have the same form as the articulation of F, and no condition is assigned to any particular term of it.

Neither contrast secures the other. An articulation of F may be realized while no contrast anywhere holds in virtue of it, and a field may bear contrasts that hold in virtue of nothing in F. A field's merely differing within itself is therefore not registration.

(R) also fixes how much individuation the account admits. There must be enough to distinguish K from K', and to take r and r' as positions of one contrast. There need not be enough to identify enduring objects or successive states of a continuing subject: r and r' are both actual positions of one contrast, neither alternatives nor stages in a history. To treat these stronger forms of individuation as already established would build into the foundation a persisting, bounded bearer—which is precisely what an account of a knowing subject must eventually explain.

6. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

6. Registration

Registration can now be defined in a restricted, structural sense.

DEFINITION OF REGISTRATION

A contrast K borne by F is registered by a contrast K' when C holds between them: K' holds at least partly in virtue of K. The field R that bears K' is then said to register K.

Registration is thus nothing over and above an instance of C, and it is always registration *of* a particular contrast: to say that R registers F without qualification is to leave the claim incomplete. An account that states a registration must therefore specify an articulation; the specification is the account's, and the registration does not depend on it. It is also an actual relation between actual contrasts, not a standing disposition of R to respond. Figure 3 depicts the structure so defined.

The definition attributes to the receiver no understanding, identification, or judgement, since none of these capacities has been introduced. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place anything "in the logical space of reasons".[7] Nor does registration move anything from F to R: the receiving contrast need neither resemble the articulation nor reproduce its organization, provided that the dependence holds. Nor, finally, does it require a representational space, or any repertoire of alternatives, in R. Registration relates contrasts that are realized, and nothing in R need stand for conditions that R does not exhibit.

Registration so defined is supplied by neither field in isolation. It comprises three distinct contributions: from the target, the realized articulation; from the receiver, the contrast between its conditions; and between them, the dependence that connects the two. Keeping these contributions distinct prevents registration from being attributed to the target merely because the target is differentiated, or to the receiver merely because its conditions differ. This does not explain the emergence of a knowing subject, but it identifies what such an explanation would have to elaborate.

What is registered, on this account, is always a difference within F, never which way F is. The core provides for registering that a differs from b; it does not provide for registering that F is a rather than b, since that would require b

7. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, is a case of registration in the present sense (Dretske 1969).

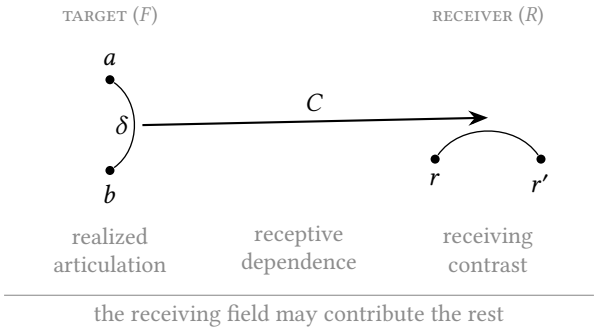


FIGURE 3. Registration. Through C, the contrast between the conditions r and r' of R holds at least partly in virtue of the contrast of the realized articulation $\langle a, \delta, b\rangle_F$. All four terms are actual. The receiving contrast is drawn with a different organization, since it need neither resemble nor reproduce the articulation, and no term is paired with a condition. The arrow runs from contrast to contrast and denotes receptive dependence only: it represents neither a transfer nor an order of occurrences, and the columns are positions in C, not regions.

to be an alternative to a rather than a term beside it. Accounts of information and knowledge built on that richer form presuppose a set of alternatives among which the actual case is singled out.[8] The core deliberately admits no such set, since it would require a repertoire of possibilities, held in F, in R, or elsewhere, that nothing in the guiding premise demands. Whether registration of which way F is can be reconstructed from contrasts between actual terms, without admitting alternatives, is left open. Such a reconstruction may also need the pairing of terms with conditions, which (C) does not include.

7. Registration and Order

The core contains no order of any kind. Its contrasts are not sequences, and the conditions of R are both actual, not successive. C is asymmetric, but the asymmetry of a relation between its positions is not an ordering of occurrences: that K' holds in virtue of K does not place either before the other.

The core therefore cannot describe the receiver as *acquiring* a contrast, *retaining* it, or *comparing* it with another, and each of these operations requires

8. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

something different.[9] Comparison requires two receiving contrasts that are both actual; whatever else it requires, it does not require order. Retention requires that one receiver be identified across distinct positions of some ordering. Acquisition requires that a position of such an ordering at which a contrast is not registered stand in an asymmetric relation to one at which it is. Only retention and acquisition require order, and neither follows from (C).

What they require is order in the loosest sense: an asymmetric relation among positions, with no requirement that any two positions be comparable, and with no metric, no linearity, and no universal index. Whether this relation must also be transitive, and so form a strict partial order, may be left open. Temporal succession is one way of realizing such an order; the order of steps in an inference or a proof is another, which is not temporal. The core is compatible with either and requires neither.

For the same reason, if receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The asymmetry of dependence and the asymmetry of an ordering play different roles in the account and should be kept notationally distinct.

8. Conclusion

Table 1 lists everything the account assumes, by kind, with what each item leaves unestablished. Only (C) is a commitment, but it is not the only thing assumed.

The reconstruction rests on one commitment and on two notions that it does not analyse. The resulting proposal is that registration, in whatever domain it is realized, requires *receptive dependence* between contrasts of actual terms, with target and receiver distinguished as positions within that dependence. The claim is one of necessity only, and the necessity is conditional: whatever is perceived, or registered in any way, satisfies these conditions. They are not sufficient for perception or for knowledge, and nothing is claimed here about aboutness by description.

The core is minimal, but it is not empty: it excludes several views of what registration presupposes. First, it sets aside three ways in which a difference might be supplied from the wrong side. A respect whose holding depends, even partly, on the specification or its notation lies outside the reconstruction;

9. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account requires only order, of which temporal order is one kind.

TABLE 1. What the account assumes, by kind, and what each item leaves unestablished.

ITEM	KIND	WHAT IT LEAVES UNESTABLISHED
The guiding premise	Premise, not argued here	That registration occurs; anything about aboutness by description.
Bearing a contrast	Notion used without analysis	Which contrasts a field bears; any capacity or possibility of the field.
Holding in virtue of	Notion used without analysis	What the dependence consists in within a domain; whether it is transitive.
(C) Receptive dependence	Commitment	Which way F is; a single registered contrast; a pairing of terms; understanding, representation, order, acquisition, or retention.
Scope of the reconstruction, with the strict asymmetry	Condition of method	That no difference is supplied by a description or by its own registration.
Actual terms only	Decision	That there are no possibilities; only that the core uses none.
(D), the lone term, (R)	Consequences	That any field bears a contrast apart from registration; that R and F are distinct in substance.
The logic of the exposition; the individuation of contrasts	Presupposed	Not examined here.

this is a matter of scope, by the condition of Section 2, not a result. A field's merely differing within itself, where its contrast does not hold even partly in virtue of the target's contrast, is not registration, by (C). And a contrast that held, even partly, in virtue of the contrast that registers it lies outside the reconstruction, again as a matter of scope (Section 4). Second, the core excludes the view that registration consists in resemblance or in term-by-term correspondence, since both are symmetric and neither supplies the dependence required by (C) (Section 4). Third, it excludes any conception of registration as a transfer from target to receiver, since the receiving contrast need neither resemble nor reproduce the articulation (Section 6). Fourth, it excludes the view that a repertoire of alternatives, whether held in the target, in the receiver, or elsewhere, is a precondition of registration, since every contrast in the core is between actual terms (Sections 3 and 6). Fifth, it excludes the view that time, or order of any kind, is such a precondition, since registration has been defined without either (Section 7). Finally, it excludes the view that a knowing subject, concepts, or representations precede registration, since none of these figures in (C) or in what follows from it. None of these exclusions denies that the structures concerned are present in particular epistemic relations. What is denied is only that they belong to the core: wherever they are needed, they must enter as further commitments, stated as such.

Five limitations should be stated. First, the guiding premise is not argued for in this paper, and the necessity claimed is no stronger than the premise. Second, the two notions on which the core rests are not analysed. The bearing of a contrast carries the whole weight of the distinction between a contrast of the target and one supplied by a description. Receptive dependence is characterized only as relating what actually holds and as asymmetric within the scope of the reconstruction, and many relations have these properties. The core is a schema, and it has not been shown here, by worked cases, that the schema is met in more than one domain. Third, registration as defined is indiscriminate: a contrast registers every contrast it holds at least partly in virtue of. Fourth, the core claims neither that the receiving field contributes anything to the registering contrast nor any bound on what it contributes; how much it may contribute belongs to an account of fidelity, at a later stage. Fifth, the account picks out the contrasts of its targets by description, and the guiding premise does not cover aboutness by description: the account uses a relation to its targets that it does not itself cover.

Five questions remain open. The first is what the dependence C consists in, and what grounds its asymmetry, in a given domain. The second is whether receptive dependence can itself be explained in terms of more elementary relations, and whether it is transitive; an arrow whose meaning is left unspecified would not answer it. The third is whether registration of which way F is can be reconstructed without admitting alternatives. The fourth is what order acquisition and retention require, and whether anything stronger than an asym-

metric relation is needed. The fifth is whether aboutness by description, of the future, and of mathematical objects is built from registered contrasts. The reconstruction ends at structural registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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